

Text Classification in NLP

CMPE346 · Group 2

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Group task

Text Classification

Metric

Accuracy & F1

Student	Task
Atakan Gül	Media Bias Detection
Ece Doğa Gül	Sexism Detection
Talha Demir	Ghost-Job Risk Classification

BiLSTM (baseline)

Bidirectional recurrent network trained from scratch — no pretraining.

BERT `bert-base-uncased` & RoBERTa `roberta-base`

Pretrained transformer encoders fine-tuned end-to-end; RoBERTa adds dynamic masking and removes the NSP objective.

Task

Classify a news sentence as *biased* or *not biased*.

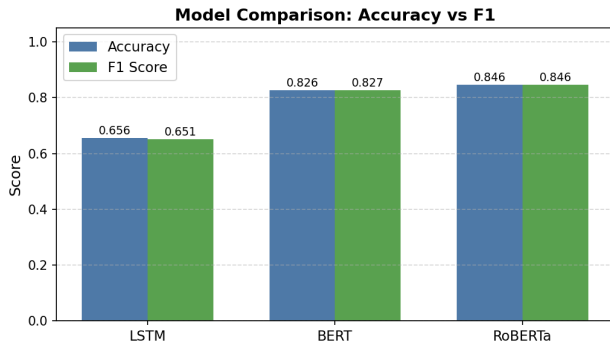
Dataset: BABE

- 4,121 expert-annotated sentences
- Diverse news outlets
- 3,121 train / 1,000 test

Atakan — Results

Model	Acc.	W-F1
BiLSTM	0.6560	0.6511
BERT	0.8260	0.8266
RoBERTa	0.8460	0.8464

- RoBERTa best: +19pp accuracy over BiLSTM
- BERT close behind
- BiLSTM struggles on subtle bias



Atakan — Prediction Examples

Sentence	True	LSTM	BERT	RoBERTa
<i>“Companies like Nike and Google made June-teenth a paid holiday.”</i>	NB	B ×	NB ✓	NB ✓
<i>“None of Biden’s Democrat counterparts challenged his claim of economic selflessness.”</i>	B	B ✓	NB ×	B ✓

NB = Not Biased B = Biased ✓ = correct × = incorrect

- Pretraining is critical: transformers capture subtle bias patterns BiLSTM cannot
- RoBERTa is the best model — 84.60% accuracy, 0.8464 weighted F1

Task

Classify a social media comment as *sexist* or *not sexist*.

Dataset: EDOS

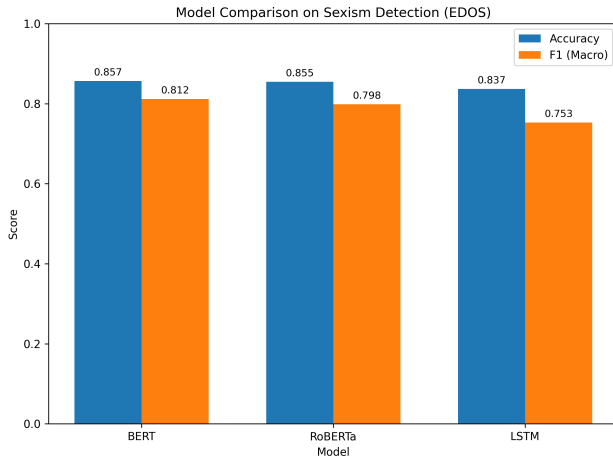
- 20,000 comments from Reddit & Gab
- SemEval-2023 Task 10
- 14,000 train / 2,000 dev / 4,000 test

Ece — Results

Model	Acc.	F1
BiLSTM	0.8365	0.7530
BERT	0.8565	0.8115
RoBERTa	0.8545	0.7981

F1 = macro F1

- BERT best across both metrics
- Transformers handle class imbalance better than BiLSTM



- Pretrained transformers outperform the BiLSTM baseline on sexism detection
- BERT is the best model — 85.65% accuracy, 0.8115 macro F1

Task

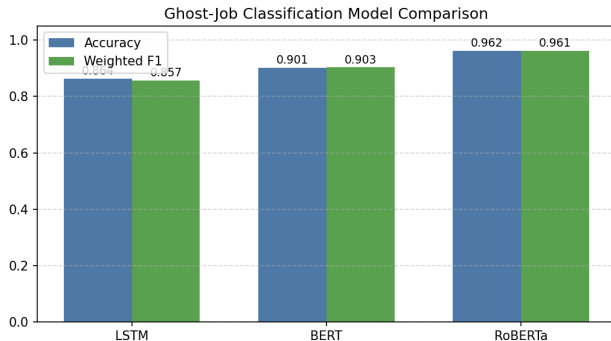
Classify job postings by visible ghost-job risk evidence.

Dataset: Zenodo 20321172

- unified_core job postings
- Evidence-based target correction
- 3:1 realistic-to-ghost downsampled splits

Model	Acc.	W-F1
BiLSTM	0.8636	0.8569
BERT	0.9015	0.9029
RoBERTa	0.9621	0.9610

- RoBERTa best overall
- BERT improves likely-ghost recall over BiLSTM



Talha — Prediction Examples

Posting signal	True	RoBERTa	Why it matters
<i>"This position is contingent upon contract award."</i>	Ghost	Ghost ✓	Explicit contingency
<i>"Join our talent community for future opportunities."</i>	Ghost	Ghost ✓	Pipeline-only wording
<i>"We are shaping the future of AI with an active engineering team."</i>	Realistic	Ghost ×	Hard phrase: harmless "future" context

Ghost = likely ghost-risk ✓ = correct × = incorrect

- Dataset correction was essential because raw ghost labels were noisy
- Labels reflect visible text signals, not verified hiring outcomes
- RoBERTa reaches 96.21% accuracy and 0.9610 weighted F1

All Results

Task	Model	Acc.	F1
Media Bias	BiLSTM	0.6560	0.6511
	BERT	0.8260	0.8266
	RoBERTa	0.8460	0.8464
Sexism Detection	BiLSTM	0.8365	0.7530
	BERT	0.8565	0.8115
	RoBERTa	0.8545	0.7981
Ghost Jobs	BiLSTM	0.8636	0.8569
	BERT	0.9015	0.9029
	RoBERTa	0.9621	0.9610

Media Bias and Ghost Jobs: weighted F1 Sexism Detection: macro F1

- Transformers consistently outperform BiLSTM — pretraining is decisive on small datasets
- The best transformer can vary by task: RoBERTa leads media bias and ghost jobs, while BERT leads sexism detection